Detecting anti-holomorphic dynamics via symbolic regression with the eml* operator

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May 2026

Abstract

We introduce $\mathrm{eml}^\star(x, y) = \exp(\overline{x}) - \ln(\overline{y})$, the anti-holomorphic extension of the EML operator $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$ of Odrzywołek (2026). We prove that $\overline{z} = 1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1))$ (Theorem 1), providing a purely algebraic representation of complex conjugation within the EML framework. Combined with symbolic regression (PySR), this yields an automatic criterion for classifying complex dynamical systems: eml* appears in the discovered equation when the underlying dynamics is anti-holomorphic and does not appear when it is holomorphic, across all tested systems. We validate this on three domains: (i) fractal iteration maps, where PySR distinguishes the Mandelbrot set ($z^2 + c$, holomorphic, eml*-free) from the Tricorn ($\overline{z}^2 + c$, anti-holomorphic, eml*-present) at machine precision (MSE $\sim 10^{-32}$); (ii) quantum state evolution, where eml* correctly identifies dissipative systems (7/7 classification); (iii) gravitational lensing ellipticities from KiDS-1000, confirming eml* detection on 5 000 real astrophysical measurements. A negative control on the weak gravitational shear profile around KiDS-1000 lenses (268 722 source-lens pairs) shows that eml* does not appear at low complexity when the underlying physics is holomorphic (standard general relativity), confirming specificity.

1 Introduction

A fundamental distinction in complex dynamics is between holomorphic maps, which preserve the complex structure, and anti-holomorphic maps, which reverse it through complex conjugation. This distinction governs whether a system conserves information (unitary evolution in quantum mechanics, conformal maps in fluid dynamics) or dissipates it (damped oscillations, measurement collapse).

Despite its importance, no automatic method existed to determine from data alone whether a governing equation is holomorphic or anti-holomorphic. Standard symbolic regression tools [Cranmer, 2023] search over real-valued operators and cannot distinguish z^2 from $\overline{z}^2$ since these coincide on the real line.

Odrzywołek [Odrzywołek, 2026] showed that a single binary operator $\mathrm{eml}(x, y) = \exp(x) - \ln(y)$ generates all elementary functions through uniform binary trees. We extend this framework with $\mathrm{eml}^\star$, the anti-holomorphic companion of eml, and show that it provides the missing piece: an intrinsic marker of anti-holomorphic structure that appears automatically in symbolic regression when the underlying dynamics requires complex conjugation.

2 The $\mathrm{eml}^\star$ operator

Definition 1 (EML operator)**.** *Following Odrzywołek [2026], the EML operator is:*

$$\mathrm{eml}(x, y) = \exp(x) - \ln(y), \quad y \neq 0. \tag{1}$$

Definition 2 ($\mathrm{eml}^\star$ operator)**.** *The anti-holomorphic extension of EML is:*

$$\mathrm{eml}^\star(x, y) = \exp(\overline{x}) - \ln(\overline{y}), \quad y \neq 0. \tag{2}$$

Note that $\mathrm{eml}^\star$ coincides with eml on $\mathbb{R}$, since $\overline{x} = x$ for $x \in \mathbb{R}$. This is not a defect but a feature: $\mathrm{eml}^\star$ can only be distinguished from eml on data with genuine complex structure.

Theorem 1 (Conjugation via $\mathrm{eml}^\star$)**.** *For all $z \in \mathbb{C}$ with $|\mathrm{Im}(z)| < \pi$:*

$$\overline{z} = 1 - \mathrm{eml}^\star(0,\ \mathrm{eml}(z, 1))\,. \tag{3}$$

Proof. We compute stepwise:

$$\mathrm{eml}(z, 1) = \exp(z) - \ln(1) = \exp(z), \tag{4}$$

$$\mathrm{eml}^\star(0, \exp(z)) = \exp(\overline{0}) - \ln(\overline{\exp(z)}) \tag{5}$$

$$= 1 - \ln(\exp(\overline{z})) \tag{6}$$

$$= 1 - \overline{z}, \tag{7}$$

where the last step uses $\ln(\exp(w)) = w$ on the principal branch ($|\mathrm{Im}(w)| < \pi$, satisfied since $\mathrm{Im}(\overline{z}) = -\mathrm{Im}(z)$). Therefore $1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1)) = 1 - (1 - \overline{z}) = \overline{z}$. □

Corollary 1 (Modulus squared)**.** $|z|^2 = z \cdot \overline{z} = z \cdot [1 - \mathrm{eml}^\star(0, \mathrm{eml}(z, 1))]$.

Corollary 2 (Density)**.** *The algebra* $\mathcal{A}$ *generated by* $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ *is dense in* $C(K, \mathbb{C})$ *for any compact* $K \subset \mathbb{C}$ *contained in a horizontal strip of height less than* 2π*, by the Stone–Weierstrass theorem for complex algebras. The algebra separates points (since* $\mathrm{eml}(z, 1) = \exp(z)$ *is injective on such strips), contains the constants, and is self-adjoint (closed under conjugation, since* $\mathrm{eml}^\star$ *provides* $\overline{z}$ *via Theorem 1).*

3 A Re-based decomposition of $\mathrm{eml}^\star$

The $\mathrm{eml}^\star$ operator was introduced in Section 2 as the natural anti-holomorphic companion of eml. A symbolic regression experiment using PySR was run with z and w as independent complex inputs sampled from the square $\{|\mathrm{Re}|, |\mathrm{Im}| \leq \pi - 0.05\}$ with $|w| \geq 0.1$, and the toolbox restricted to $\{+, -, \times, /, \sin, \cos, \exp, \log, \mathrm{my\_real}, \mathrm{my\_ima}$ (no `my_conj`, no eml, no $\mathrm{eml}^\star$). On 2 000 training points, PySR rediscovered the following identity at MSE $= 4.22 \times 10^{-31}$ (machine precision for complex128):

$$\boxed{\mathrm{eml}^\star(z, w) = \mathrm{Re}\big(2\exp(z) - \ln(w^2)\big) + \ln(w) - \exp(z)} \tag{8}$$

An independent verification on $N = 10\,000$ points (different seed) confirmed MSE $= 3.17 \times 10^{-31}$ and maximum absolute error 7.11×10^{-15}.

Algebraic proof. For any $u \in \mathbb{C}$, the elementary identity $\overline{u} = 2\,\mathrm{Re}(u) - u$ holds. Applying it to $u = \exp(z)$ and using $\overline{\exp(z)} = \exp(\overline{z})$ yields

$$\exp(\overline{z}) = 2\,\mathrm{Re}(\exp(z)) - \exp(z). \qquad \text{(Lemma A)}$$

Applying it to $u = \ln(w)$, and using $\overline{\ln(w)} = \ln(\overline{w})$ on the principal branch ($\arg(w) \in (-\pi, \pi)$), yields

$$\ln(\overline{w}) = 2\,\mathrm{Re}(\ln(w)) - \ln(w). \qquad \text{(Lemma B)}$$

Substituting Lemma A and Lemma B into the definition $\mathrm{eml}^\star(z, w) = \exp(\overline{z}) - \ln(\overline{w})$ gives equation (8). $\square$

Structural consequence. Equation (8) expresses $\mathrm{eml}^\star$ using only Re, exp, ln, and arithmetic operations. Since $\{\mathrm{eml}, 1\}$ generates exp and ln (Odrzywołek, 2026), it follows that $\{\mathrm{eml}, \mathrm{Re}, 1\}$ generates $\mathrm{eml}^\star$. Combined with the reverse direction (Re is recoverable from $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ via $\mathrm{Re}(z) = (z + \overline{z})/2$ and Theorem 1), we obtain:

Theorem 2 (Equivalence of minimal non-holomorphic extensions)**.** *The operator sets* $\{\mathrm{eml}, \mathrm{eml}^\star, 1\}$ *and* $\{\mathrm{eml}, \mathrm{Re}, 1\}$ *have identical expressive power over* $C(K, \mathbb{C})$ *for every compact* $K \subset \mathbb{C}$ *contained in a horizontal strip of height less than* 2π *and disjoint from* $\{w = 0\}$*.*

Corollary 3 (Generalized completeness)**.** *Let* ϕ *be any non-holomorphic primitive in* $\{\mathrm{Re}, \mathrm{Im}, \bar{\cdot}, \mathrm{eml}^\star\}$*. The set* $\{\mathrm{eml}, \phi, 1\}$ *is dense in* $C(K, \mathbb{C})$ *for every compact* K *satisfying the branch condition above.*

Discussion. Theorem 2 relativizes the apparent uniqueness of $\mathrm{eml}^\star$ as the natural anti-holomorphic extension of eml. Any single non-holomorphic primitive added to $\{\mathrm{eml}, 1\}$ achieves $C(K, \mathbb{C})$-density. The $\mathrm{eml}^\star$ formulation retains structural appeal (it preserves the binary, exp-minus-log signature of eml), while the Re formulation is more minimal in the standard analytic toolkit.

The empirical rediscovery of equation (8) by an unconstrained symbolic regression search demonstrates that the Re-based decomposition is not merely formally valid but *algorithmically accessible*: a generic evolutionary search over elementary operators converges to it within minutes when given enough expressive flexibility. Implementation and reproducibility scripts are available at `https://github.com/antparis/oxieml-star` (commit `f03a46e`).

4 Method

4.1 Symbolic regression with complex operators

We use PySR [Cranmer, 2023] (v1.5.10) with Julia backend, configured for complex arithmetic:

- **Binary operators:** $+, -, \times, \div, \mathrm{eml}, \mathrm{eml}^\star$
- **Unary operators:** $\cos, \sin, \exp, \log, \mathrm{my\_conj}(z) = \overline{z}$, $\mathrm{my\_real}(z) = \mathrm{Re}(z)$, $\mathrm{my\_imag}(z) = \mathrm{Im}(z)$, $\mathrm{my\_abs2}(z) = z\overline{z}$
- **Precision:** `Complex{Float64}`
- **Reproducibility:** `deterministic=True, parallelism="serial", random_state=42`

4.2 Detection criterion

A system is classified as *anti-holomorphic* if the best equation found by PySR contains any of: `emlstar`, `my_conj`, `my_abs2`, or `my_imag`. These all involve complex conjugation.

4.3 Why $\mathrm{eml}^\star$ is necessary

Standard symbolic regression over $\{+, -, \times, \div, \exp, \sin, \cos, \log\}$ cannot distinguish $f(z) = z^2$ from $g(z) = \overline{z}^2$, because all standard operators are holomorphic. Adding $\mathrm{eml}^\star$ to the operator set is the minimal extension that makes anti-holomorphic functions expressible. By Corollary 2, $\{\mathrm{eml}, \mathrm{eml}^\star\}$ suffices to approximate any continuous complex function.

5 Results

5.1 Fractal iteration maps

We generate 200 random points $z \in [-1, 1]^2 \subset \mathbb{C}$ (seed 42) and compute one iteration of three fractal maps with $c = -0.7 + 0.27015i$:

Table 1: Fractal maps: PySR discovers the exact iteration rule and correctly identifies the holomorphic/anti-holomorphic nature.

Fractal	True rule	PySR equation	$\mathrm{eml}^\star$?	MSE
Mandelbrot	$z^2 + c$	$(x_0 \cdot x_0) - (0.7 - 0.27i)$	No	1.86×10^{-32}
Tricorn	$\overline{z}^2 + c$	$\mathrm{my\_conj}((x_0 \cdot x_0) - (0.7 + 0.27i))$	**Yes**	2.02×10^{-32}
Burning Ship	$(\lvert\mathrm{Re}\rvert + i\lvert\mathrm{Im}\rvert)^2 + c$	approximation	**Yes**	1.39×10^{-2}

The Mandelbrot and Tricorn results are at machine precision. PySR recovers the exact algebraic form and uses my_conj (equivalent to $\mathrm{eml}^\star$ via Theorem 1) for the Tricorn but not for the Mandelbrot. This is the cleanest possible test: data is natively complex, no encoding artefact is possible, and the only difference between the two maps is one conjugation.

5.2 Quantum state evolution

We generate time series $\psi(t) \to \psi(t+dt)$ for three quantum systems ($\omega = 1.0$, $dt = 0.1$, 300 time steps):

All seven systems are classified correctly. $\mathrm{eml}^\star$ appears when the physics requires non-trivial conjugation: dissipation ($|U| < 1$) or measurement ($|z|^2$ with varying z). For the EM plane wave, $|E + iB|^2 = 1$ is constant (the wave is on the unit circle), so PySR discovers the constant directly without conjugation. It does not appear for unitary or conservative systems.

Table 2: Quantum evolution: PySR finds the exact evolution operator and identifies dissipative systems via eml*.

System	$\lvert U\rvert$	eml*?	MSE	Physics
Larmor precession	1.0000	No	5.24×10^{-32}	Unitary
Damped oscillation	0.9900	**Yes**	1.69×10^{-32}	Dissipative
Rabi oscillation	1.0000	No	2.79×10^{-31}	Unitary

Table 3: Multi-domain physics: additional natively complex systems.

System	Observable	eml*?	MSE
Impedance (RLC circuit)	$\lvert Z(\omega)\rvert^2$	**Yes**	2.48×10^{-31}
Bell state (Born rule)	$\lvert\psi\rvert^2$	**Yes**	1.74×10^{-33}
ECG analytic signal	$\lvert z(t)\rvert^2$	**Yes**	5.58×10^{-34}
EM plane wave	$\lvert E + iB\rvert^2 = 1$ (const.)	No	1.97×10^{-33}

5.3 Astrophysical validation: KiDS-1000

As a first application to real astrophysical data, we use 5 000 galaxy ellipticities from the KiDS-DR4 DEIMOS catalog [Giblin et al., 2021]: each galaxy has a measured shape $\gamma = e_1 + i\, e_2 \in \mathbb{C}$. We test whether PySR recovers $|\gamma|^2 = \gamma \cdot \overline{\gamma}$:

$$\text{PySR finds: my\_abs2}(x_0), \quad \text{MSE} = 2.54 \times 10^{-34}, \quad \text{eml}^\star = \textbf{Yes}. \tag{9}$$

This confirms that eml* operates correctly on real astrophysical measurements with native complex structure. While the test is a sanity check ($|\gamma|^2$ is known to require conjugation), it is the first application of eml* to observational data.

6 Negative control: weak gravitational shear profile

The previous tests show eml* appearing on systems where anti-holomorphic structure is expected from theory. A stronger test of specificity is to apply the same machinery to data where anti-holomorphic structure is *not* expected and verify that eml* does not appear.

We use the KiDS-1000 SOM-gold catalog [Giblin et al., 2021], comprising 21 262 011 source galaxies with reliable shape and photometric redshift measurements over 1006 $\deg^2$, and the 268 high-quality strong-lens candidates of Petrillo et al. [2019], Li et al. [2020, 2021]. After standard selection cuts (`fitclass` $=0$, `MASK` $=0$, `weight` >0, `Z_B` >0.3) we retain 17 450 042 sources. Pairing every source within 10 arcmin of each lens yields 293 880 (lens, source) pairs spread across 181/268 lenses with at least one matched source. Of these, 268 722 pairs fall within the analysis range $r \in [0.5, 10]$ arcmin used for stacking.

For each pair, we compute the lens–source angular separation $\Delta z = (\mathrm{RA}_s - \mathrm{RA}_\ell) + i(\mathrm{Dec}_s - \mathrm{Dec}_\ell)$ and the rotated source ellipticity $\gamma_{\mathrm{rot}} = \gamma \exp(-2i\varphi)$, where $\gamma = e_1 + ie_2$ and $\varphi = \arg(\Delta z)$. We stack pairs in 20 logarithmic bins of $r = |\Delta z|$ between 0.5 and 10 arcmin, using `lensfit` inverse-variance weights:

$$\langle \gamma_{\mathrm{rot}} \rangle(r) \;=\; \frac{\sum_k w_k \, \gamma_{\mathrm{rot},k}}{\sum_k w_k}. \tag{10}$$

All 20 bins have effective stacking weight $N_{\mathrm{eff}} > 100$ and are retained.

Anti-circularity. The rotation factor $\exp(-2i\varphi) = (\overline{\Delta z}/|\Delta z|)^2$ contains an implicit conjugation. If Δz or φ were passed to PySR, the system could trivially reconstruct γ_{rot} via complex conjugation, giving a circular result. To prevent this, PySR receives only $r = |\Delta z|$ as a single real-valued input (cast to `complex128` with zero imaginary part), with $\langle \gamma_{\mathrm{rot}} \rangle(r)$ as the complex target. This is verified in code by asserting $\mathrm{Im}(X) = 0$ before fitting.

Result. PySR with the full eml*-enabled toolbox is run on the 20-bin stacked profile. The Pareto front shows a clear holomorphic plateau at low complexity: the formula $(c_1 + ic_2)/(x_0 + c_3)$ at complexity 5 reaches MSE $= 5.7 \times 10^{-5}$, a factor 2.4 improvement over the constant baseline. This is the expected $1/r$ shear scaling from standard general relativity. Complexity-11 reaches MSE $= 3.9 \times 10^{-5}$ with a purely holomorphic formula and *beats* the only anti-holomorphic candidate at complexity 10 (MSE 4.5×10^{-5}, score 0.16). The eml* operator does appear at complexity ≥ 14, but with a marginal score per added node ($\lesssim 0.03$) and within the regime where PySR can interpolate noise on 20 binned points.

Interpretation. This is the contrapositive of the positive tests of Section 5. On a real astrophysical signal that is well described by holomorphic general relativity, eml* is *not* structurally required: PySR prefers simpler holomorphic formulas at low complexity and only resorts to eml* for cosmetic high-complexity fits. This is the behavior expected of a specific anti-holomorphic detector, not a free-floating fit parameter.

7 Discussion

7.1 What eml* detects

Across all validated tests, eml* appears when the dynamics involves complex conjugation and does not appear when it does not:

- Anti-holomorphic maps ($\bar{z}^2 + c$)
- Dissipative evolution ($|U| < 1$, requiring $|z|^2 = z\bar{z}$)
- Born rule measurements ($P = |\psi|^2$)

It does not appear for holomorphic, unitary, or energy-conserving systems.

7.2 Limitations

Branch cut restriction. Theorem 1 requires $|\mathrm{Im}(z)| < \pi$ for the principal branch of the logarithm. All data in this work satisfies this condition; extensions to $|\mathrm{Im}(z)| \geq \pi$ would require branch-tracking.

Real-valued data. eml* is structurally identical to eml on $\mathbb{R}$. The method is only informative on data with genuine complex structure. This is a mathematical fact, not a limitation of the implementation.

Neural world model. We attempted to build a JEPA-style neural network (complex-valued, with separate holomorphic and anti-holomorphic pathways). In a supervised setting, the model achieves 88% classification accuracy, but without supervision, the learned gate does not differentiate the two pathways. The root cause is that $|\bar{z}| = |z|$: a gate that receives only magnitudes cannot distinguish holomorphic from anti-holomorphic dynamics. This remains an open problem.

7.3 Coordinate representations

We tested eml* on the Schwarzschild metric in two formulations: the standard Hilbert form (which crosses the horizon $r = 2M$ and produces complex values) yields eml* = True, while the original Schwarzschild form (which avoids the interior) yields eml* = False. This is a result about coordinate representations, not about physical reality: the coordinate singularity at $r = 2M$ has been understood since Kruskal (1960).

8 Conclusion

We have shown that the eml* operator, combined with symbolic regression, provides an automatic, interpretable criterion for detecting anti-holomorphic structure in complex dynamical systems. The method is validated on fractal maps (machine precision), quantum evolution (7/7 classification), and real astrophysical data (KiDS-1000), with rigorous negative controls confirming specificity.

The eml* framework opens several directions: (i) application to natively complex astrophysical data (gravitational lensing shear fields, interferometric visibilities); (ii) integration as a structural prior in neural world models; (iii) extension to quaternionic or Clifford-algebraic settings.

All code and data are available at `https://github.com/antparis/eml_star`.

Acknowledgments

The author thanks A. Odrzywołek for the EML operator framework, and the KiDS collaboration for public data access. Symbolic regression was performed with PySR [Cranmer, 2023]. This work was conducted independently without institutional affiliation or funding.

References

Cranmer, M. (2023). Interpretable machine learning for science with PySR and SymbolicRegression.jl. *arXiv:2305.01582*.

Giblin, B., et al. (2021). KiDS-1000 catalogue: Weak gravitational lensing shear measurements. *Astronomy & Astrophysics*, 645, A105.

Petrillo, C. E., et al. (2019). LinKS: discovering galaxy-scale strong lenses in the Kilo-Degree Survey using convolutional neural networks. *Monthly Notices of the Royal Astronomical Society*, 484(3), 3879–3896.

Li, R., Napolitano, N. R., Tortora, C., et al. (2020). New high-quality strong lens candidates with deep learning in the Kilo-Degree Survey. *The Astrophysical Journal*, 899(1), 30.

Li, R., Napolitano, N. R., Spiniello, C., et al. (2021). High-quality strong lens candidates in the final Kilo-Degree Survey footprint. *The Astrophysical Journal*, 923(1), 16.

Odrzywołek, A. (2026). All elementary functions from a single binary operator. *arXiv:2603.21852*.